

PROJECT DOCUMENTATION

SUPER STORE SALES

Power BI Specialist

Prepared by:

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Data Analytics Project | April 2026

Course:

Power BI Specialist

Dataset:

- Primary Source: superstore_sales.csv

Provided by the Digital Egypt Pioneers Initiative (DEPI) . It includes a dataset of 9,800 rows and 18 columns containing transactional information regarding order and shipping dates, customer demographics, regional logistics, products, and sales.

- Kaggle Superstore Dataset

Used to enrich the primary data with critical financial metrics. This source contributed the Quantity, Discount, and Profit columns, allowing for a complete analysis of profit margins and unit volume

Objective:

Analyze retail sales data to identify profitability drivers, discount impact, and regional performance gaps — and deliver actionable recommendations to optimize revenue and margin.

Tools Used:

Microsoft Power BI · Power Query. Excel · PowerPoint

SECTION 1: PROJECT PLANNING & MANAGEMENT

1.1 Project Proposal

This project analyzes four years of retail sales data (2015–2018) from a US-based superstore to uncover profitability drivers, discount impact, regional performance gaps, and customer behavior patterns. The outcome is a set of actionable business recommendations supported by interactive Power BI dashboards.

1.1.1 Problem Statement

Revenue growth alone does not guarantee business sustainability. Despite posting \$2.26M in total sales over four years, the store's overall profit margin stands at only 12.34%. Three product sub-categories generate net losses, and aggressive discounting is eroding margins across multiple regions. Without data-driven insight, the business risks continued margin degradation.

1.1.2 Project Objectives

- Evaluate overall revenue performance and year-over-year sales trends.
- Identify which product categories and sub-categories are profitable and which are loss-making.
- Quantify the relationship between discount levels and profit outcomes.
- Assess regional and state-level performance gaps.
- Deliver a two-quarter sales forecast using Power BI's time-series model.
- Produce a prioritized, actionable set of business recommendations.

1.1.3 Scope

The project covers all order-level data available in the Superstore Sales Dataset. Analysis is limited to the United States market across four regions (West, East, Central, South) from January 2015 to December 2018. External factors (macroeconomic conditions, competitor activity) are outside the scope.

1.2 Project Plan — Timeline & Milestones

The project was structured into five phases across an eight-week delivery cycle. The table below summarizes the key milestones, owners, and delivery status.

Task	Responsible	Start	End	Status
Dataset acquisition & review	All members	Week 1	Week 1	Complete
Data cleaning & prep	All members	Week 1	Week 2	Complete
KPI definition & metrics	All members	Week 2	Week 2	Complete
Sales & category analysis	Mohamed Ibrahim	Week 2	Week 3	Complete
Shipping & logistics page	Saeed Rabie	Week 3	Week 3	Complete
Customer analysis page	Ahmed Abd El-Salam	Week 3	Week 4	Complete
Product analysis page	Bishoy Saad	Week 4	Week 4	Complete
Regional geography page	Mohamed Fawzy	Week 4	Week 4	Complete
Strategic insights & forecast	Jehad Mahmoud	Week 5	Week 5	Complete
Report writing — first draft	All members	Week 5	Week 6	Complete
Peer review & revision	All members	Week 6	Week 6	Complete
Presentation preparation	All members	Week 7	Week 7	Complete
Final submission	All members	Week 8	Week 8	Complete

1.3 Task Assignment & Team Roles

The following table defines each team member's primary responsibility area within the project, their specific deliverables, and contribution type.

Member	Role	Primary Deliverable	Section
Mohamed Ibrahim	Project Lead / Overview	Executive Dashboard (Page 1) · Final Report	Sales KPIs & Trends
Saeed RABIE	Logistics Analyst	Shipping & Logistics Dashboard (Page 2)	Shipping Analysis
Ahmed Abd El-Salam	Customer Analyst	Customer Analysis Dashboard (Page 3)	Customer Segmentation
Bishoy Saad	Product Analyst	Product Performance Dashboard (Page 4)	Sub-Category Deep Dive
Mohamed Fawzy	Geographic Analyst	Regional Geography Dashboard (Page 5)	Regional Performance
Jehad Mahmoud	Strategy & Insights Analyst	Strategic Insights Dashboard (Page 6A & 6B)	Forecasting & Strategy

1.4 Risk Assessment & Mitigation Plan

The following risks were identified during project planning. Each risk is rated by likelihood and impact, with a mitigation strategy defined.

Risk	Likelihood	Impact	Mitigation Strategy
Data quality issues / missing values	Low	High	Run data validation checks before analysis; document any gaps found.
Conflicting analytical findings between members	Medium	Medium	Peer review each section; reconcile using source data as ground truth.
Time overrun on documentation phase	Medium	Low	Set documentation deadline 1 week before final submission.
Visualization misinterpretation by audience	Low	Medium	Add clear chart titles, data labels, and written summaries on all pages.

1.5 Key Performance Indicators (KPIs)

The following KPIs were defined to measure both business performance (from the dataset) and project delivery quality.

1.5.1 Business KPIs

KPI	Definition	Value	Target / Benchmark
Total Revenue	Sum of all sales transactions	\$2,261,537	Year-over-Year (YoY) growth
Total Profit	Net profit across all transactions	\$278,979	> \$250,000
Profit Margin %	$\text{Profit} \div \text{Sales} \times 100$	12.34%	> 15% (industry avg)
Avg Order Value (AOV)	$\text{Total revenue} \div \text{number of unique orders}$	\$459.48	> \$400
Avg Discount Rate	Mean discount applied across all transactions	15.68%	< 12% (recommended)
Customer Retention	$\text{Repeat buyers} \div \text{total customers} \times 100$	98.4%	> 80%
Total Orders	Count of unique order IDs	4,922	Grow YoY
Units Sold	Sum of all quantities across all orders	37,143	Grow YoY

SECTION 2: LITERATURE REVIEW

2.1 Feedback & Evaluation

This section documents the academic and peer feedback received during the project lifecycle, lecturer assessments, and the criteria used to evaluate project quality.

2.1.1 Methodological Justification

The analytical approach draws on established retail analytics methodologies. Key literature references supporting the project's framework include:

- Profit margin analysis as a primary performance indicator .
- Geographic profitability mapping as a tool for regional resource allocation decisions.
- Time-series forecasting validation using confidence intervals — 95% .

2.2 Suggested Improvements

Based on peer and instructor review, the following improvement areas were identified

I. Navigation & User Interface

- Designed a vertical navigation panel on the left side of the canvas to mimic a modern application interface.
- Implemented page navigation buttons with "On Hover" effects to improve user interactivity.
- Configured button actions to provide seamless transitions between the Main Dashboard and individual member pages.
- Synchronized the navigation bar across all report pages to ensure a consistent user experience.

II. Executive Dashboard Development

- Constructed a Main Dashboard to serve as a high-level executive summary for the entire team.
- Utilized aggregate KPI cards to display total performance metrics and team-wide averages.
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III. Individual Team Analysis

- Developed six dedicated report pages, one for each specific team member.
- Applied page-level filters to isolate and display data exclusive to the respective team member.
- Standardized the layout across all six pages to facilitate easy comparison of individual metrics.
- Hidden detail pages from the bottom tab bar to force navigation through the primary side panel.

IV. DAX Calculations & Logic

- Authored custom DAX measures to calculate core performance indicators (KPIs) across the dataset.
- Created dynamic DAX-based page titles that automatically update based on the selected team member.

V. Data Visualization & Formatting

- Enabled data labels on all visuals to show exact values directly inside the charts.
- Applied conditional formatting to highlight values that meet or exceed project targets.
- Standardized number formatting to ensure all currency and percentage values are consistent throughout the report.

2.3 Final Grading Criteria

The project was evaluated across the following assessment dimensions:

Assessment Area	Weight	Key Criteria
Documentation Quality	25%	Completeness, clarity, structure, and professional formatting of all written deliverables.
Data Cleaning & Preparation	15%	Validation steps, derived field creation, and justification of transformation decisions.
Analytical Depth	20%	Quality of insights, root-cause identification, and use of advanced analytics (clustering, forecast).
Dashboard Design & Usability	20%	Visual clarity, interactivity, appropriate chart selection, and storytelling coherence.
Recommendations & Strategy	10%	Practicality, specificity, and business relevance of recommended actions.
Presentation & Communication	10%	Clarity of verbal explanation, slide quality, and ability to answer questions.

SECTION 3: REQUIREMENTS GATHERING

3.1 Stakeholder Analysis

The following stakeholders were identified as the primary audience and beneficiaries of the project outputs. Each stakeholder group has distinct interests and requirements from the analysis.

Stakeholder	Role	Interest in Project	Key Requirement
Store Management	Primary decision-maker	Understand overall financial health and growth opportunities	Executive summary dashboard with top KPIs
Sales Team	Revenue generators	Identify which products and regions to prioritize	Product and regional profitability breakdowns
Finance / Accounting	Profit & cost control	Quantify margin erosion and discount policy impact	Discount analysis and profit margin KPIs

3.2 User Stories & Use Cases

The following user stories define how each stakeholder interacts with the project outputs.

3.2.1 User Stories

As a...	I want to...	So that I can...
Store Manager	See total sales, profit, and margin on a single dashboard	Quickly assess overall business health without manual calculation.
Finance Officer	Filter profit by category and see discount vs profit scatter	Identify which product lines are destroying margin and act on them.
Sales Manager	View regional sales vs profit comparison by state	Reallocate sales effort to high-margin regions like California.
Category Manager	See sub-category profit waterfall chart	Immediately identify Tables and Bookcases as loss-making and escalate.
Logistics Manager	Compare shipping mode by delivery days and profit margin	Decide whether to promote First Class to high-margin product lines.
Marketing Manager	View customer segmentation by value and recency	Focus retention budget on Strategic MVP customers and re-engage lapsed ones.
Executive	See two-quarter sales forecast with confidence interval	Plan inventory and staffing levels ahead of Q4 peak season.

3.3 Functional Requirements

The following functional requirements define the specific capabilities the project deliverables must provide.

3.3.1 Dashboard Requirements

1. The dashboard must display total Sales, Profit, Profit Margin %, Total Orders, and Average Order Value as headline KPIs.
2. The system must allow filtering by Region, Category, Segment, and Year using interactive slicers.
3. Sales and profit trends must be visualized over time at both yearly and monthly granularity.
4. Sub-category profitability must be displayed as a waterfall or comparative bar chart, clearly distinguishing gains from losses.
5. A geographic map must show sales and profitability at state level, with color coding for profit vs. loss.
6. A two-quarter sales forecast must be generated using time-series analysis with a 95% confidence interval.
7. A discount reduction What-If scenario must simulate profit changes based on user-defined discount adjustments.

3.3.2 Reporting Requirements

8. A written final report must cover all 6 analysis dimensions with findings and recommendations.
9. All charts and tables in the report must include data labels and clear titles.
10. The report must include an executive summary, methodology section, and conclusion.

3.4 Non-Functional Requirements

3.4.1 Performance

- Dashboard pages must load within 5 seconds on standard hardware.
- Filtering operations must respond within 2 seconds with the full 9,800-row dataset.

3.4.2 Usability

- All visualizations must be interpretable without prior data analytics training.
- Chart titles and axis labels must be present on every visual.

3.4.3 Reliability

- All KPI values must be reproducible from the source dataset with no manual overrides.
- Data cleaning steps must be documented so results can be independently verified.

3.4.4 Security & Data Integrity

- The source dataset must not be modified — all transformations must be applied in Power Query preserving the original file.

SECTION 4: SYSTEM ANALYSIS & DESIGN

4.1 Problem Statement & Objectives

The Superstore dataset reveals a business generating strong revenue (\$2.26M over 4 years) but with a profit margin of only 12.34% — well below the 15–20% retail industry benchmark. Three sub-categories (Tables, Bookcases, Supplies) are generating net losses despite meaningful sales volumes. The Central region delivers only 7.7% profit margin compared to 14.8% in the West. Discounts as high as 80% are being applied without profit controls.

The analytical system must answer four core business questions:

- Where is sales & profit being made — and where is it being destroyed?
- What is the relationship between discounting and profitability?
- Which customer segments and geographic areas represent the highest opportunity?
- What will sales performance look like over the next two quarters?

4.1.1 Use Case Diagram & Descriptions

System Actors

Actor	Type	Description
Store Manager	Primary User	Views KPI dashboards and executes strategic decisions.
Analyst (Team)	Primary User	Builds, maintains, and interprets all analytical outputs.
Finance Officer	Secondary User	Reviews discount and margin analysis for cost control.
Power BI Engine	System Actor	Processes DAX, renders visuals, and runs forecasting model.
Data Source	External Actor	Excel/CSV Superstore dataset; provides raw transaction data.

Key Use Cases

Use Case	Actor	Precondition	Main Flow	Outcome
View KPI Dashboard	Manager	Dashboard is published	Open Power BI → select page 1 → view KPIs	Manager sees headline \$2.26M / \$279K / 12.34%
Filter by Region	Analyst	Slicers configured	Click region slicer → all visuals update	Region-specific profit and sales displayed
Identify Loss-Making Products	Finance Officer	Sub-category page open	View waterfall chart → identify red bars	Tables and Bookcases flagged for review
Run Discount Simulation	Manager	What-If parameter set	Adjust slider → view projected profit	Manager sees +\$141K profit from -5% discount
View Sales Forecast	Executive	Forecast page open	View line chart with forecast band	Two-quarter outlook with 95% CI displayed

Use Case	Actor	Precondition	Main Flow	Outcome
Export Report	Analyst	Report finalized	Export Power BI report → compile Word doc	Final documentation package created

4.1.2 Software Architecture

The project employs a layered analytics architecture with three tiers:

Architecture Tiers

Tier	Components	Technology	Role
Data Layer	Raw dataset, cleaned dataset, derived metrics	Excel / CSV, Python (pandas)	Stores and prepares all raw and transformed data.
Processing Layer	Power Query (ETL), DAX measures, Python scripts	Power BI Desktop, Python	Cleans, transforms, and computes all analytical metrics.
Presentation Layer	6 dashboard pages, written reports, presentations	Power BI, Word, PowerPoint	Delivers insights to stakeholders in accessible formats.

4.2 Database Design & Data Modeling

4.2.1 Entity-Relationship Overview

The Superstore dataset is a flat transactional file. For analysis purposes, it can be conceptualized as a star schema with the following logical entities:

Entity	Primary Key	Key Attributes	Relationship
Orders	Order ID	Order Date, Ship Date, Ship Mode, Customer ID, Postal Code	One order → many order lines
Customers	Customer ID	Customer Name, Segment	One customer → many orders
Products	Product ID	Product Name, Category, Sub-Category	One product → many order lines
Geography	Postal Code	City, State, Region, Country	One location → many orders
Order Lines	Row ID	Order ID, Product ID, Sales, Quantity, Discount, Profit (derived)	Fact table — many per order

4.2.2 Logical & Physical Schema

Field	Type	Source / Derived	Notes
Row ID	Integer	Source	Unique identifier per order line.
Order ID	String	Source	Groups multiple line items into one order.
Order Date	Date	Source	Formatted as date in Power Query.
Ship Date	Date	Source	Used to compute Shipping Lag.
Shipping Lag	Integer	Derived	Ship Date – Order Date (days). Created in Power Query.
Customer ID	String	Source	Links to Customer entity.
Segment	String	Source	Consumer / Corporate / Home Office.
Region	String	Source	West / East / Central / South.
Category	String	Source	Technology / Furniture / Office Supplies.
Sub-Category	String	Source	17 unique sub-categories.
Sales	Decimal	Source	Revenue per line item (USD).
Quantity	Integer	Source	Units sold per line item.
Discount	Decimal	Source	Discount rate (0.0 to 0.8).
Profit	Decimal	Source	Net profit per line item (can be negative).
Profit Margin %	Decimal	Derived	Profit ÷ Sales × 100. DAX measure.
Year	Integer	Derived	Extracted from Order Date. Used for trend analysis.
Month-Year	String	Derived	MM-YYYY format. Used for monthly trend charts.

4.3 Data Flow & System Behavior

4.3.1 Data Flow Diagram (DFD) — Context Level

The context-level DFD defines the boundaries of the analytical system and its interaction with external entities.

Context DFD Description

External Inputs: Raw Superstore dataset (Excel/CSV). System Boundary: Power BI analytics environment. External Outputs: (1) Interactive dashboards → Business stakeholders. (2) Written reports → Academic assessors. (3) Presentation slides → Presentation audience. Data stores within the system boundary: Cleaned dataset, DAX measures model, Forecast output.

DFD — Level 1 (Process Decomposition)

Process ID	Process Name	Input Data	Output Data
P1	Data Ingestion	Raw Excel file	Loaded Power BI dataset
P2	Data Cleaning	Loaded dataset	Cleaned, typed, null-free dataset
P3	Metric Computation	Cleaned dataset	DAX KPIs, derived columns
P4	Forecasting	Monthly sales time-series	2-quarter forecast with confidence band
P5	Visualization	Metrics, clusters, forecast	6 interactive dashboard pages
P6	Reporting	Visualizations + analyst findings	Written report, slides, infographic

4.3.2 Sequence Diagram — Dashboard Interaction

The following sequence describes how a user interaction with the regional filter propagates through the system:

1. Power BI filter engine applies the selected context to all DAX measures on the page.
2. Total Sales measure recomputes: `CALCULATE(SUM(Sales), Region = 'Central')` → \$492K.
3. Profit Margin measure recomputes → 7.7%.

4.3.3 Activity Diagram — Analytical Workflow

The analytical workflow follows a sequential process with a validation gate before each output phase:

- Start → Load raw dataset into Power BI.
- Validate → Check for null values and data type errors in Power Query.
- Transform → Apply type standardization; create derived columns (Profit Margin %, Year, Lag).
- Define KPIs → Write DAX measures for all headline metrics.
- Build visuals → Construct dashboard pages per assigned team member.
- Peer review → Each member reviews at least one other member's dashboard page.
- Revise → Incorporate feedback; ensure visual consistency across pages.
- Export → Publish to Power BI Service; export screenshots for reports.
- Document → Compile written report, slides, and infographic.
- End → Submit final package.

4.3.4 Class Diagram — Analytical Components

The following logical classes represent the major components of the analytical system:

Class	Attributes	Methods / Calculations
Transaction	OrderID, CustomerID, ProductID, Date, Sales, Profit, Discount, Qty	profitMargin, shippingLag, discountBand
Customer	CustomerID, Name, Segment, Region, ClusterLabel	lifetimeValue, riskScore, segmentTier
Product	ProductID, Name, Category, SubCategory	totalProfit, avgDiscount, profitRank
Geography	City, State, Region	regionalMargin, stateRank, profitGap
Dashboard	Title, Visuals, Slicers, Measures	render, filter, export
ForecastModel	TimeSeries	predict, computeCI, detectSeasonality

4.4 UI/UX Design & Prototyping

4.4.1 Dashboard Layout Principles

- Each dashboard page follows a consistent layout: headline KPIs at the top, main charts in the center, and a written insight or callout at the bottom.
- The global filter panel (Region, Category, Segment, Year) appears consistently on every page.
- Page navigation tabs at the bottom of every page allow instant access to any section.
- A maximum of 5 visuals per page ensures cognitive clarity and avoids information overload.

4.4.2 Accessibility Considerations

- All charts include data labels to ensure information is accessible without relying solely on color.
- Tooltips are enabled on all chart visuals to provide additional context on hover.

4.5 System Deployment & Integration

4.5.1 Technology Stack

Layer	Technology	Version	Role
Data Storage	Microsoft Excel (.xlsx)	Microsoft 365	Raw dataset storage and initial review.
ETL / Prep	Power Query (M Language)	Power BI Desktop	Data cleaning, type casting, derived columns.
Analytics Engine	DAX (Data Analysis Expressions)	Power BI Desktop	KPI calculation, filtering, aggregation.
Forecasting	Power BI Analytics Forecast	Built-in	Time-series 2-quarter outlook (95% CI).

Layer	Technology	Version	Role
Visualization	Microsoft Power BI Desktop	Latest	Interactive dashboard creation and publishing.
Reporting	Microsoft Word / PowerPoint	Microsoft 365	Written report and presentation slides.
Distribution	Power BI Service / PDF / PPTX	Cloud + Local	Stakeholder delivery and academic submission.

4.5.2 Deployment Diagram

The project deployment architecture involves three environments:

- Local Development Environment: Power BI Desktop installed on each team member's machine.. Excel dataset stored on shared drive.
- Collaboration Environment: Shared cloud folder (Google Drive) for dataset and report file version control. Power BI Desktop files (.pbix) version-managed by file naming convention.
- Presentation / Distribution Environment: Power BI Service (free tier) for published dashboard sharing. PDF export for report submission. PowerPoint file for presentation delivery.

4.5.3 Component Diagram

The analytical system consists of the following components and their dependencies:

Component	Depends On	Provides To
Raw Dataset (Excel)	None	Power Query ETL
Power Query (ETL)	Raw Dataset	Cleaned Data Model
Data Model (Power BI)	Power Query output	DAX Measures, Visuals
DAX Measures	Data Model	All KPI visuals
Power BI Forecast Module	Monthly sales time-series	Forecast visual on Page 6
Dashboard Pages (1–6)	Data Model + DAX + Python + Forecast	Business users, academic assessors
Written Report	Dashboard screenshots + analyst notes	Academic submission
Presentation Slides	Written report + dashboard screenshots	Presentation audience

4.6 Additional Deliverables

4.6.1 DAX Measures Documentation

The following DAX measures were created to support the analytical model:

Measure Name	DAX Expression
Total Revenue	SUM('Orders'[Sales])
Total Profit	SUM('Orders'[Profit])
Profit Margin %	DIVIDE(SUM('Orders'[Profit]), SUM('Orders'[Sales]), 0) * 100
Total Orders	DISTINCTCOUNT('Orders'[Order ID])
Avg Order Value	DIVIDE([Total Revenue], [Total Orders], 0)
Avg Discount	AVERAGE('Orders'[Discount])
Total Customers	DISTINCTCOUNT('Orders'[Customer ID])
Repeat Customers	COUNTROWS(FILTER(CustomerTable, [order_count] > 1))

4.6.2 Testing & Validation Plan

Test Type	Description	Method	Pass Criterion
Data completeness	Verify 0 null values across all 21 columns	Power Query → Column quality check	0% null / 0% error across all columns
KPI accuracy	Cross-validate DAX total revenue vs Excel SUM	Manual Excel calculation	DAX result matches Excel to 2 decimal places
Filter integrity	Confirm slicers update all visuals correctly	Click each slicer; verify all charts	All 5+ visuals update on each filter change
Forecast validation	Back-test 2017 forecast vs 2017 actuals	Hold out 2017; forecast from 2015–16	Forecast within 10% of actuals
Cross-browser render	Confirm dashboard renders in Edge, Chrome, Firefox	Open Power BI Service in each	No visual distortion or loading errors

4.6.3 Deployment Strategy

- Primary submission: Power BI .pbix file submitted alongside PDF exports of all 6 dashboard pages.
- Backup copies: All files stored on shared cloud drive (Google Drive) with version history enabled.

Super Store Sales — Complete Project Documentation

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